

CS673 Product Update

Project Name: GradAid

Group Name: Admissions Alchemists

Github URL: <https://github.com/yuezhang0594/GradAid/tree/refactor>

Team Profile (Unchanged)

Yue Zhang:

Yue is developing the frontend, ensuring a smooth and intuitive user experience. He also works on backend API integration and user authentication, enabling secure sign-in and seamless communication between the UI and database. He ensures application security by implementing input validation on all website components.

Nitin Krishna Bojji:

Works with LLM and agentic AI to integrate these technologies into our platform, automating the creation of SOPs and LORs for master's students. His focus is on enhancing performance, scalability, and security to ensure a responsive and reliable user experience.

Joseph Rissman:

Joseph's focus for this project is developing and maintaining the database, integrating it with the other components to ensure a smooth flow of data. He is working to ensure user data security by implementing role access control measures at the database level.

Product Vision (Unchanged)

For international students planning to attend graduate school in the US who need to draft documentation required for their applications. GradAid is an AI Driven Service Provider that provides students with freedom to be creative and tailor stories to their individual needs at a low-cost. Unlike educational consultancies, our product is available to users at any time and at a lower cost.

Personas (Unchanged)

Persona 1 - Li Lei

Li Lei, age 24, is a Chinese graduate student completing his MS degree in Biomedical Science. Originally from China, he earned his bachelor's degree in Chemistry before pursuing his master's studies. Li is passionate about scientific research and is determined to pursue a PhD in the United States to further his academic career. Despite his strong research background, he struggles with academic writing in English and is unfamiliar with the specific expectations of U.S. graduate applications. Li hopes to craft compelling Statements of Purpose (SOPs) and Letters of Recommendation (LORs) that effectively showcase his research achievements and align with his PhD aspirations. He views GradAid as an essential tool to generate professional, grammatically accurate documents tailored to his needs while helping him refine his tone and clarity for a natural-sounding application.

Persona 2 - Astha Jain

Astha Jain, a 28-year-old from Jaipur, India, comes from a large family and was the first to attend college. She pursued her passion for medicine at Mahatma Gandhi University of Medical Sciences and Technology, earning a Bachelor of Medicine and Bachelor of Surgery (MBBS) degree with high marks. Although she initially aspired to become a surgeon, she faced challenges finding employment immediately after graduation. Relocating to New Delhi in search of opportunities, she worked as a server to support herself while continuing her job search. Eventually, she secured a position as a junior doctor at a family medicine practice, where she worked for two years until the practice closed due to financial difficulties. Following this setback, she accepted a nursing position at Jeewan Mala Hospital in New Delhi. This career detour, coupled with the demanding hours of her new role, has prompted her to re-evaluate her career path. She now plans to pursue a graduate degree in a growing and more lucrative field, such as computer science or finance. Having saved enough money to study in the United States, she hopes that a graduate degree from a top American university will open doors to high-paying jobs, enabling her to support her aging parents, whose ability to work is diminishing.

Unsure of which specific field to pursue, Astha is finding it difficult to identify suitable programs. Even when she finds a seemingly appropriate program, she struggles to articulate her diverse experiences in a compelling Statement of Purpose (SOP). While acknowledging that her career journey has been unconventional, she recognizes the valuable lessons and transferable skills she has acquired along the way. She believes that an educational consultant could be instrumental in navigating the application process by understanding her unique story, her personal journey, and her future ambitions.

Persona 3 - Carlos Mendoza

Carlos Mendoza, age 26, is a software engineer from Mexico City with four years of experience in backend development. He earned his bachelor's degree in Computer Science from UNAM and has worked at a mid-sized tech company. Carlos has always been passionate about AI and

machine learning and wants to pivot into research, but his current job focuses more on software development rather than deep AI work.

He plans to apply for an MS in Artificial Intelligence in the United States to gain the necessary academic background to transition into AI research roles. However, he is unfamiliar with the research-heavy SOPs required for graduate applications and has limited experience writing formal academic documents in English. He is also unsure how to best frame his work experience to demonstrate its relevance to AI research. He hopes GradAid will help him craft a strong SOP that highlights his technical background while showcasing his research potential in AI.

Scenarios (Unchanged)

Scenario 1 - Li Lei

Li Lei, a 24-year-old Chinese graduate student completing his MS in Biomedical Science, is preparing to apply for PhD programs in the United States. With a BS in Chemistry and extensive research experience, he is eager to continue his academic journey in biomedical innovation. However, he faces significant challenges in crafting a compelling application due to his limited English proficiency and unfamiliarity with the tone, structure, and expectations of U.S. graduate admissions. Writing a strong Statement of Purpose (SOP) and securing well-crafted Letters of Recommendation (LORs) feels overwhelming, as he struggles to articulate his research achievements and career goals in a way that resonates with admissions committees.

To overcome these challenges, Li turns to GradAid, which helps him generate well-structured, grammatically accurate SOPs and LORs tailored to his background and aspirations. By inputting key details about his academic journey and research experience, he receives personalized drafts that he can refine using GradAid's tone adjustment and clarity-enhancing features. The platform also provides research-focused suggestions to ensure his documents align with the expectations of specific PhD programs. With GradAid's support, Li gains confidence in his application, knowing that his materials effectively highlight his strengths and are presented in a professional, natural-sounding manner.

Scenario 2 - Astha Jain

Astha is determined to pursue a graduate degree in the US, but the application process feels overwhelming. Remembering a friend's recommendation, she opens her browser and navigates to GradAid.com. Astha clicks the "Get Started" button and begins interacting with GradAid's AI chatbot.

GradAid asks her a series of questions: her background, academic achievements, work experience, career aspirations, and preferred fields of study. Astha appreciates how GradAid's conversational interface makes it easy to share her story, even the less conventional parts. She explains her medical background, her shift to nursing, and her desire to transition into a more lucrative and stable career to support her parents. Based on Astha's input, GradAid's AI

analyzes her profile and generates a list of potential graduate programs in computer science and finance at universities across the US. The recommendations are tailored to her academic record, work history, and career goals. Astha is impressed by the breadth and relevance of the suggestions.

Astha selects a few programs that pique her interest. She knows the Statement of Purpose (SOP) is crucial, and it's where she's been struggling the most. Thankfully, GradAid offers an SOP brainstorming tool. GradAid's AI generates a series of prompts and potential themes for her SOP, suggesting ways to connect her seemingly disparate experiences into a cohesive narrative. Astha finds this incredibly helpful, as she's able to recognize the transferable skills she's gained, like problem-solving, communication, and adaptability.

Scenario 3 -Carlos Mendoza

Carlos has started drafting his Statement of Purpose (SOP) for MS programs in AI, but he's struggling to connect his industry experience in backend development to his passion for machine learning. Unsure of how to frame his background, he logs into GradAid and inputs details about his education, work experience, and research interests.

GradAid analyzes his profile and suggests ways to highlight transferable skills from his software engineering background, such as his experience working with large datasets, optimization algorithms, and distributed computing. The AI-generated SOP draft weaves his technical experience into a compelling narrative, emphasizing how his work has naturally led him toward AI research. The platform also offers AI-powered editing tools to refine the language, ensuring his SOP is both professional and persuasive.

Scenario 4 - Carlos Mendoza

Carlos has shortlisted five MS programs in AI, each with slightly different focuses—some emphasize research, while others are more industry-oriented. He wants to tailor his SOP for each program but doesn't know how to efficiently modify his content.

Using GradAid's customization feature, he selects the target programs and receives program-specific suggestions for refining his SOP. The tool highlights key faculty members and research labs he should mention, recommends how to emphasize certain skills based on each program's focus, and adjusts the wording to align with the expectations of research vs. industry-focused programs. With GradAid's help, Carlos creates tailored versions of his SOP, increasing his chances of securing admission.

Product Requirements (Functional Features) (Unchanged)

Must Have – minimal for release, required for a useful product

Feature	User Story	Persona or Scenario
---------	------------	---------------------

Interactive form to collect user information and store in a database.	As an aspiring graduate student, I need to provide information about myself through an interactive form.	This relates to all personas and scenarios, requiring the process to be tailored to each individual's unique experience and skills.
User authentication.	As a multi-device user, I need to log in so that my activity and history are saved and synchronized across all devices.	This does not relate to one specific persona or scenario described here, but this is a basic feature for web applications.
Customized SOP generation, specific to university requirements.	As an applicant applying to many graduate programs, I need automated document creation based on my university choice so that I can focus more on my application strategy.	This relates to Astha's scenario as she requires application materials to be created related to a different range of programs.
Customized SOP generation, specific to user information..	As a mid-career professional seeking a graduate degree abroad, I need to have my diverse background and skillset reflected in my SOP.	This relates to Astha's scenario as she is struggling to relate her background in medicine to her career goals.
Cross-platform support.	As a global applicant with limited internet access, I need to interact with GradAid in a web browser on my phone.	This does not relate to a persona or scenario described here, but the target users of GradAid may not have access to a desktop or laptop computer.
User authentication and data security (RBAC, OAuth, etc.)	As a privacy-conscious user, I need data encryption and security measures so that my personal information is protected from unauthorized access and potential leaks.	This does not relate to a persona or scenario described here, but this is a common concern for web applications.

Want to Have – improves on a required feature or additional functionality

Feature	User Story	Persona or Scenario
Document management system.	As an applicant needing to update my application materials, I want to quickly find my relevant documents.	The scenarios described here are users first interacting with GradAid, but a returning user would have this concern.

Document formatting tools.	As an applicant preparing documents for various universities, I want to specify the format of downloaded documents.	This relates to all personas as they apply to different universities. Each program has different requirements for document submission.
Application tracking system.	As an applicant managing multiple university applications, I want a system for tracking application deadlines so that I can ensure timely submissions and avoid missing critical dates	This relates to Li's persona because he is unfamiliar with the US application process and would appreciate the assistance.
Application tracking system.	As someone that struggles with organization and planning, I want a feature for tracking the progress of my applications so that I can stay informed about their status and manage any required follow-up actions.	This relates to Astha's persona because she is applying to so many different types of programs she can get overwhelmed and wants to have help keeping track of them all.
Program search.	As a prospective graduate student exploring various academic options, I want to interactively search through university programs and save them to a personalized list.	This relates to Carlos' persona because he is interested in pursuing AI research, so he would want to see all of the AI-focused programs available.

Nice to Have – things to do if there is time

Feature	User Story	Persona or Scenario
Speech-to-text and text-to-speech.	As an applicant with accessibility needs, I want GradAid to have speech recognition and speech generation capabilities.	This does not relate to a persona or scenario described here, but the project should strive to be accessible if possible.
UI/UX design and testing.	As a user with limited computer literacy, I want the GradAid website to be inviting with a modern design.	This relates to Astha's persona because she does not have the same level of computer skills as the other personas.

AI-power document updates.	As someone who has gained a new qualification, I want to be able to update my SOPs and LORs so that my new accomplishment is included.	This relates to Carlos' persona because he is learning more about AI, so if he earns a certification or completes a course he would want to have that reflected in his application documents.
University shortlisting.	As a prospective graduate student, I want to receive a shortlist of universities based on my experience and accomplishments.	This relates to Astha's scenario as she wants to have a list of programs and universities in her fields of interest that she has a good chance of being accepted to with her background.

Architecture (Updated)

The GradAid architecture is designed to prioritize maintainability, scalability, and flexibility, achieved through a modular and component-based approach. The decision to refactor the architecture, moving from Supabase to Convex, was driven by the need to address growing project complexity and enhance the integration between the frontend and backend. This change allows for a more streamlined development process and improved data handling capabilities. The system comprises several key components, each with a specific responsibility.

The frontend, built with React, leverages Vite as the build tool for efficient development and utilizes Shadcn and Tailwind CSS for UI components, ensuring a modern and consistent design. The frontend consists of several modules: a graduate program module, allowing the user to search for programs, save them to their profile, then begin an application to selected programs; the SOP/LOR Editor, providing a rich-text editing experience; the User Dashboard, displaying saved documents and application insights; Auth & Profile, handling user authentication and profile management; and various components that facilitate data fetching from Convex. The backend, now built with Convex, handles data storage and retrieval, replacing Drizzle, Supabase, and PyPI. User authentication is handled by Clerk, providing secure user management. It includes functions for SOP/LOR generation, user management, and API endpoints. Convex offers a built-in API, simplifying communication with the frontend. Finally, the LLM Engine, powered by LangChain, consists of Prompt Templates, AI Processing for interacting with LLMs, and embeddings for retrieval-augmented generation (RAG).

Managing dependencies and minimizing coupling between these components remains crucial for maintainability. GradAid achieves this through several strategies. Frontend-Backend Integration is streamlined by Convex's built-in API, eliminating the need for a separate REST API layer. This allows for efficient data fetching and real-time updates. Clerk and Convex communicate through JWT and webhooks to manage user authentication and data synchronization. This ensures users can only access data they have permission to view, so personal information remains secure. A microservice-like approach isolates the LLM Engine

from the core backend logic, enabling experimentation with different AI models and libraries. Convex provides a flexible data handling system, simplifying database operations and migrations. A modular codebase within the backend, with functions grouped into modules, continues to reduce interdependencies and promote code reusability.

GradAid employs a structured approach to development and updates. The application architecture is written in TypeScript to improve compatibility between components, enhance error handling, and aid in debugging. React hooks are used for efficient component updates. The frontend is deployed using Netlify, while the backend and database functionalities are handled by Convex.

The updated architecture, particularly the transition to Convex, provides several benefits. It simplifies data management and API interactions, leading to more efficient development. The use of TypeScript enhances code quality and maintainability. The modular design and component-based approach continue to promote scalability and flexibility, allowing for easier addition of new features and adaptation to changing requirements.

Technical Risks (Updated)

The Admissions Alchemists team continues to navigate the challenges inherent in developing a web-centric application with limited prior web development experience. While the team has made significant strides in acquiring new skills, particularly in React, Convex, and UI development with Shadcn and Tailwind CSS, the need for continuous learning remains a key consideration. The team effectively utilized collaborative learning strategies such as pair programming and knowledge-sharing sessions to accelerate this process. For example, Joseph provided guidance on React hooks, Convex queries, and TypeScript best practices, while the team worked together to resolve environment setup and data handling issues. To mitigate this ongoing risk, the team will maintain its commitment to proactive learning, leveraging online resources and seeking guidance from experienced mentors as needed.

The team's decision to adopt React, Convex, and related technologies presented an initial risk due to the team's unfamiliarity. However, the refactoring process itself has proven to be a valuable learning experience, enhancing the team's understanding of these tools and their integration. The team's collaborative approach, characterized by regular progress reviews and focused meetings, has been instrumental in overcoming this challenge. For instance, the team successfully migrated from Supabase to Convex, addressed API querying issues, and established a more robust project structure. While the team has gained basic proficiency, staying up-to-date with best practices and the evolving landscape of React and Convex development remains an ongoing effort, especially given Convex's rapid development cycle.

Time constraints, stemming from team members' diverse commitments, remain a persistent challenge. The team's strategy of dividing the project into short, focused iterations with clearly defined milestones has proven effective in managing this risk. By prioritizing key functionalities and adhering to deadlines, the team has maintained a steady pace of progress. The agile mindset adopted by the team, emphasizing flexibility and adaptability, has also been crucial in

responding to evolving requirements and unforeseen challenges. For example, the team adjusted its development plan to accommodate the Convex migration and address critical bugs. Regular workload assessments and adjustments will continue to be essential to ensure timely delivery.

Adding to the complexity, several team members have faced health challenges during this sprint, impacting their ability to contribute consistently. The team is adapting by providing support and adjusting workloads to accommodate these unforeseen circumstances. Open communication and flexible task management are crucial to minimizing the impact of these health-related disruptions on the project's progress.

Effective communication and collaboration have been paramount in mitigating these risks. The established centralized communication channel and regular meetings have facilitated seamless information exchange and coordination. The team's culture of open communication and mutual support has fostered a collaborative environment, empowering team members to contribute effectively. This strong emphasis on communication will remain crucial as the team tackles the remaining development tasks and prepares for deployment. Specifically the in-person coding session and preparation for the final demo and presentation.

Overall, the Admissions Alchemists team has demonstrated a strong capacity to adapt and overcome technical challenges. The team's commitment to continuous learning, collaboration, and agile methodologies has been instrumental in navigating the project's complexities. While risks remain, including those related to team health, the team is well-positioned to deliver a successful and impactful web application.

Product Demonstration (Updated)

GradAid is currently not deployed to a production environment. Please fork the GitHub repo, run `npm install` then `npm run dev` to launch the project on localhost.